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MULTIMEDIA UNIVERSITY

FINAL EXAMINATION

TRIMESTER OCTOBER/ NOVEMBER 2025 (TERM ID 2530)

CDS1114 – INTRODUCTION TO DIGITAL SYSTEMS

(All sections / Groups)

27 FEBRUARY 2026

9.00 a.m - 11.00 a.m

(2 Hours)

INSTRUCTIONS TO STUDENTS

1. This question paper consists of 11 pages including cover with 5 Questions only.
2. Answer **ALL** questions. The total mark for this paper is 50 marks and the distribution of the marks for each question is given.
3. Please print all your answers in this booklet.

APPENDIX**Boolean Law:**

$$\begin{aligned}A + B &= B + A \\A \cdot B &= B \cdot A \\A + (B + C) &= (A + B) + C\end{aligned}$$

$$\begin{aligned}A \cdot (B \cdot C) &= (A \cdot B) \cdot C \\A(B + C) &= AB + AC\end{aligned}$$

Boolean Rules:

$$\begin{aligned}A + 0 &= A \\A + 1 &= 1 \\A \cdot 0 &= 0 \\A \cdot 1 &= A \\A + A &= A \\A + \bar{A} &= 1\end{aligned}$$

$$\begin{aligned}A \cdot A &= A \\A \cdot \bar{A} &= 0 \\ \overline{\bar{A}} &= A \\A + AB &= A \\A + \bar{A}B &= A + B \\(A + B)(A + C) &= A + BC\end{aligned}$$

De Morgan's theorem:

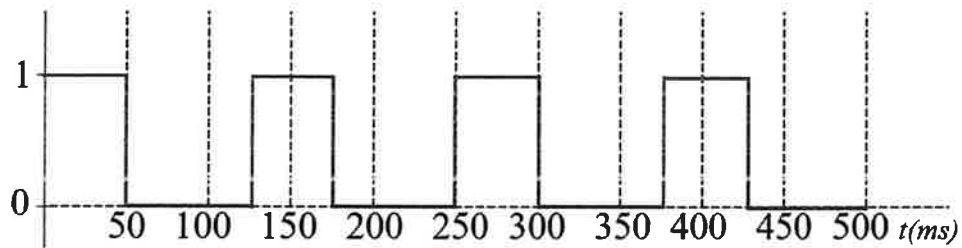
$$\overline{A \cdot B} = \bar{A} + \bar{B}$$

$$\overline{A + B} = \bar{A} \bar{B}$$

Continued

QUESTION 1

- a) Given a timing diagram is shown in Figure Q1 (a), calculate the value of period, time width and duty cycle. [3 marks]



- b) Compute the arithmetic expression below and report the final result in binary notation. Include a complete step-by-step solution.

i. $01011_2 \div 01_2$

[1 Mark]

ii. $63_8 + 1A_{16}$.

[2 Marks]

Continued

- c) Perform the calculation of $-41 - 64$ using 8-bits signed 2's complement binary numbers. Provide your final answer in decimal and hexadecimal. [4 Marks]

QUESTION 2

- a) Simplify the expression $Z = \bar{A}B(C + A) + \bar{A}\bar{C} + A\bar{B}\bar{C}$ using Boolean Algebra and De Morgan's Theorems. [5 Marks]

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- b) Explain the equivalency of X and Y in Figure 2(b) using a timing diagram with A = 0110, and B = 1100. [5 Marks]

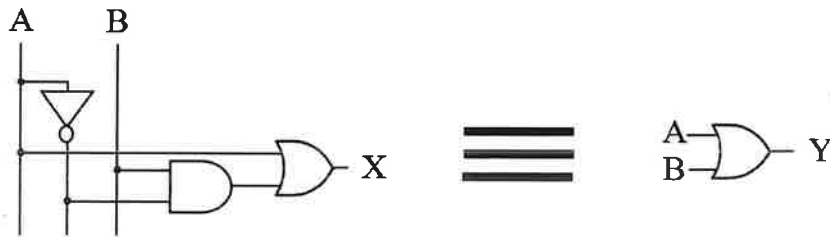
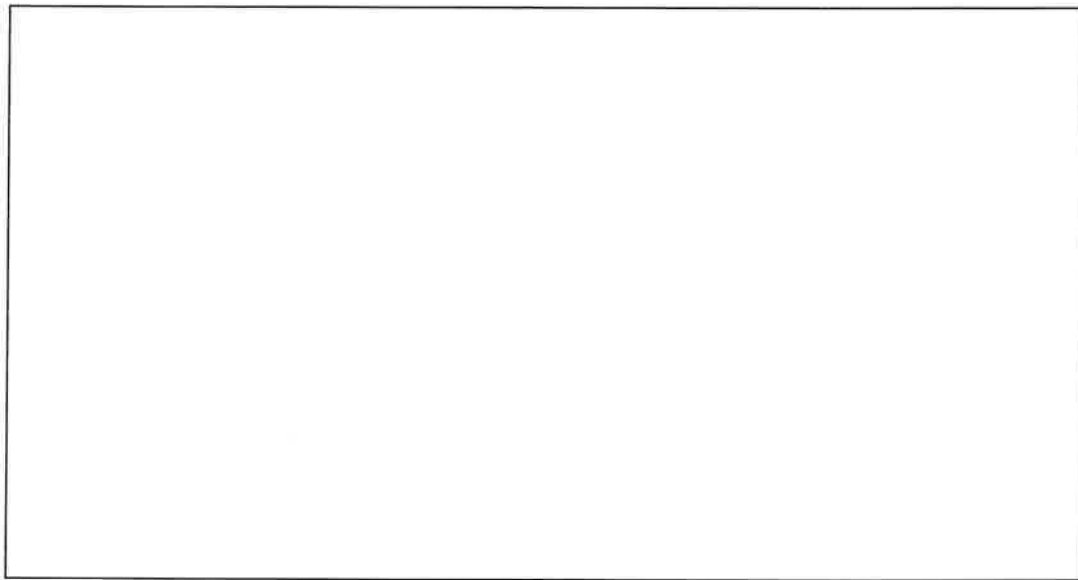


Figure 2(b)



QUESTION 3

- a) Referring to the truth table shown below, answer the following questions.

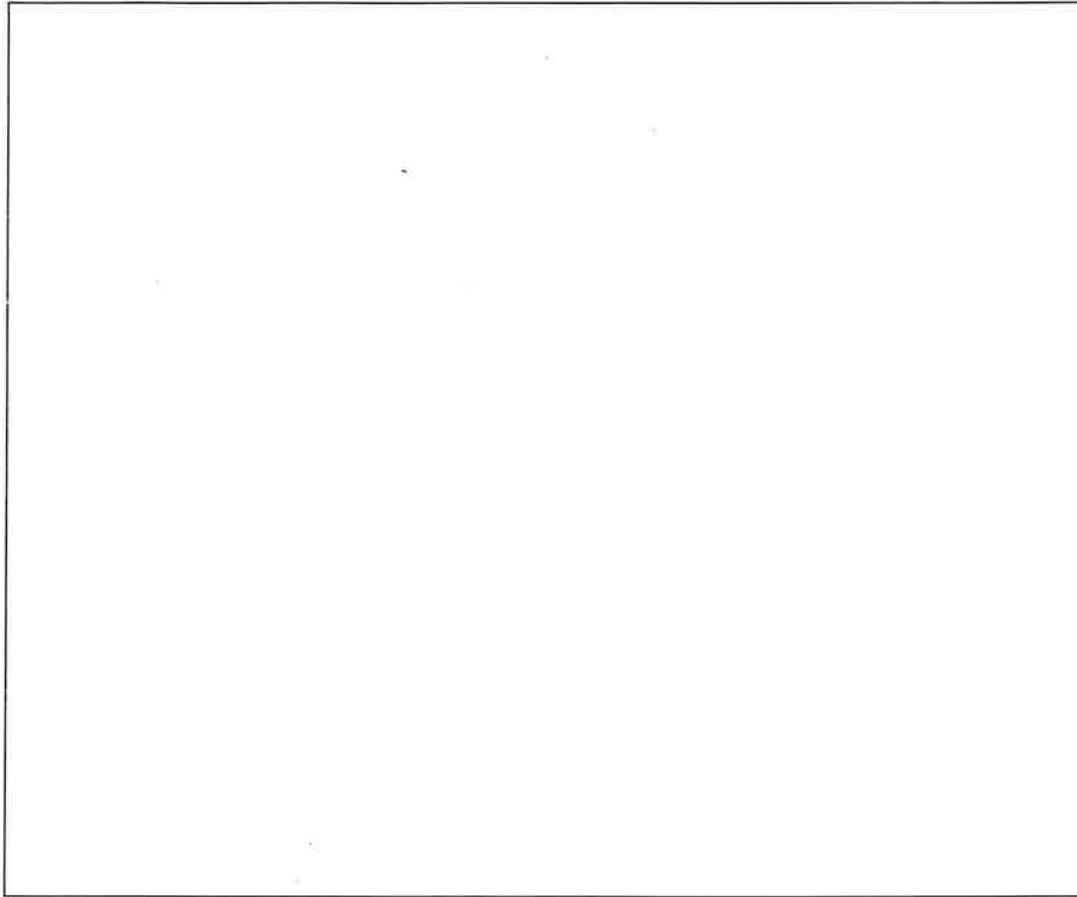
Input				Output
A	B	C	D	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1

Continued

0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

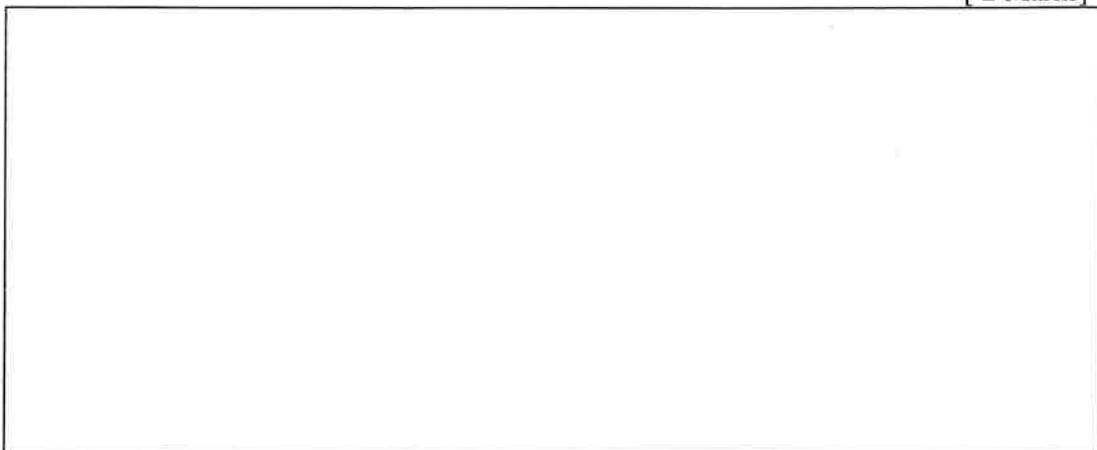
- i. Derive the Boolean expression of G in standard Product of Sums (POS) format for the given truth table. [2 Marks]
- ii. Simplify the expression in (i) using K-MAP in POS form. [5 Marks]
- iii. Construct a simplified logic circuit diagram based on the expression in (ii). [3 Marks]

Continued

**QUESTION 4**

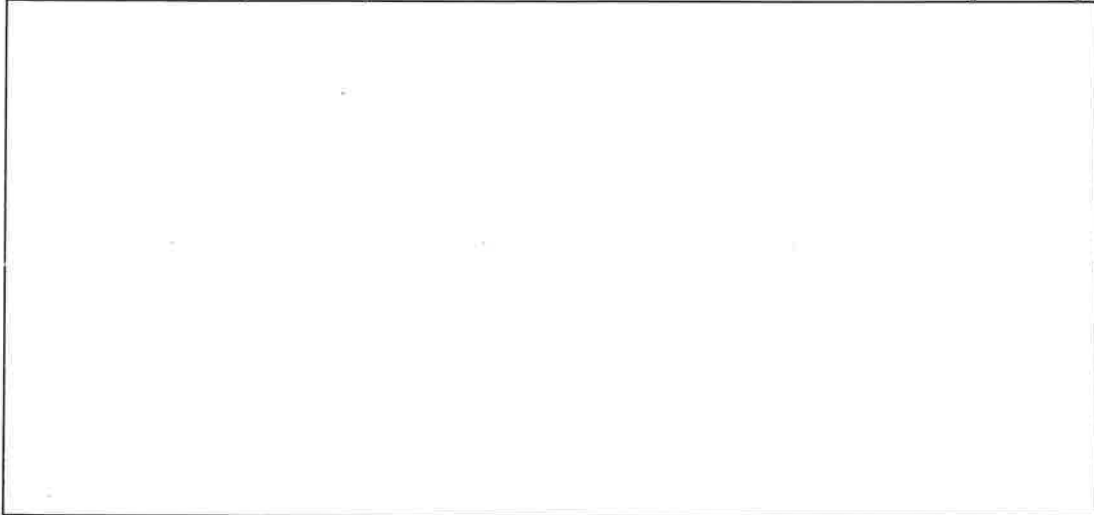
- a) Perform $1011 + 1110$ with a 4-bit parallel adder consist of only full-adders.

[2 Marks]

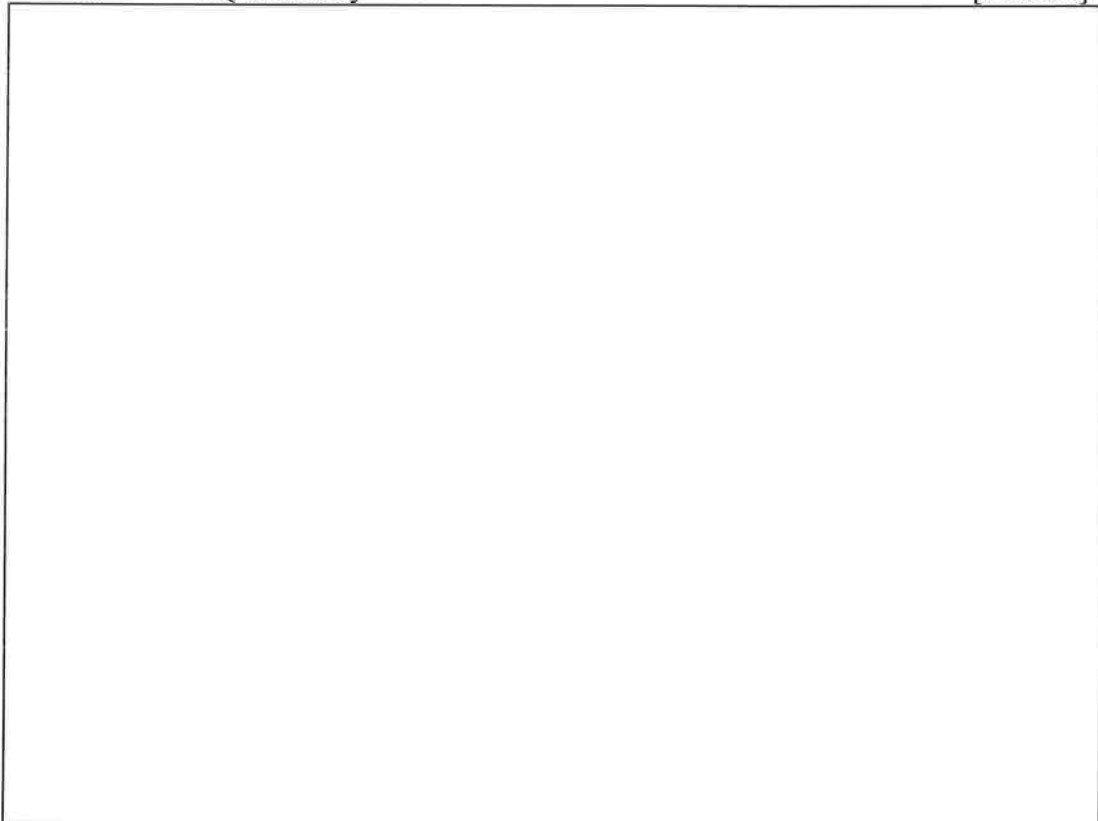


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- b) Given input S is 1001 1011 and input R is 0010 1010. Provide the Q for Active-Low S'-R' Latch. Q is initially low. [4 Marks]



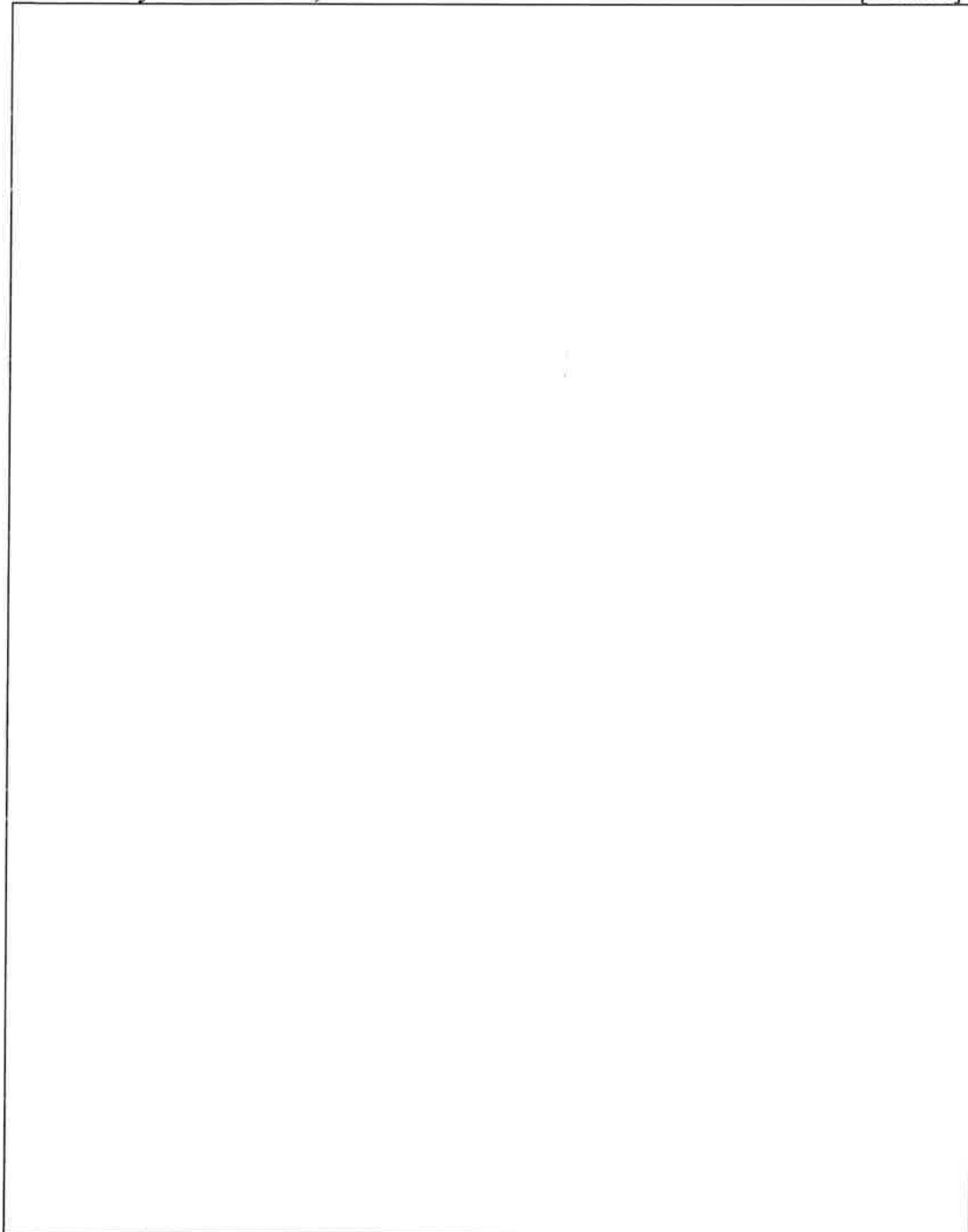
- c) Draw the Q for a Positive Going Transition (PGT) J-K flip-flop that has a clock speed rated at 200Hz. The input is fed every 5ms, and the input for J is 1011 0011 and K is 0010 1010. Q is initially low. [4 Marks]

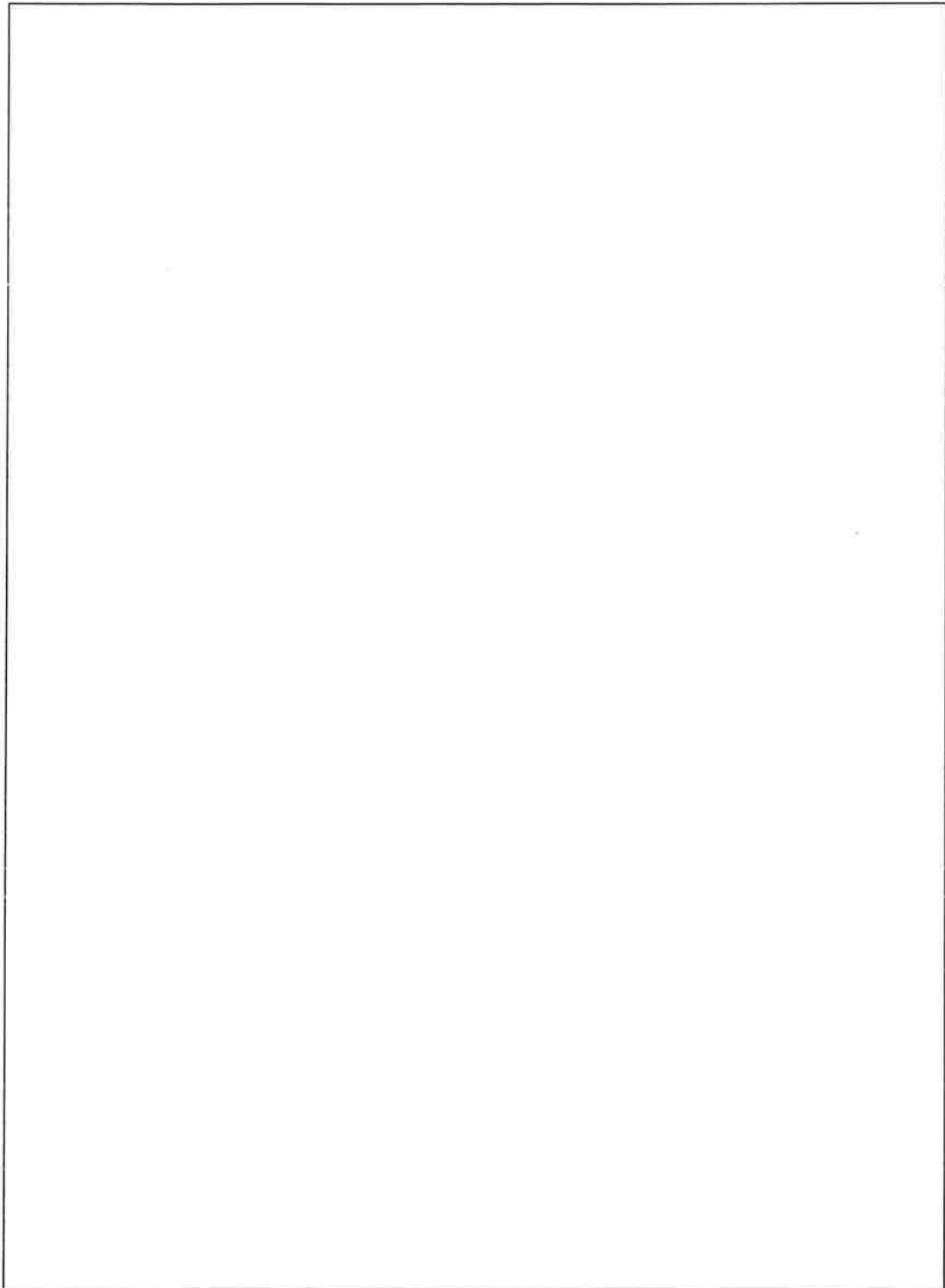


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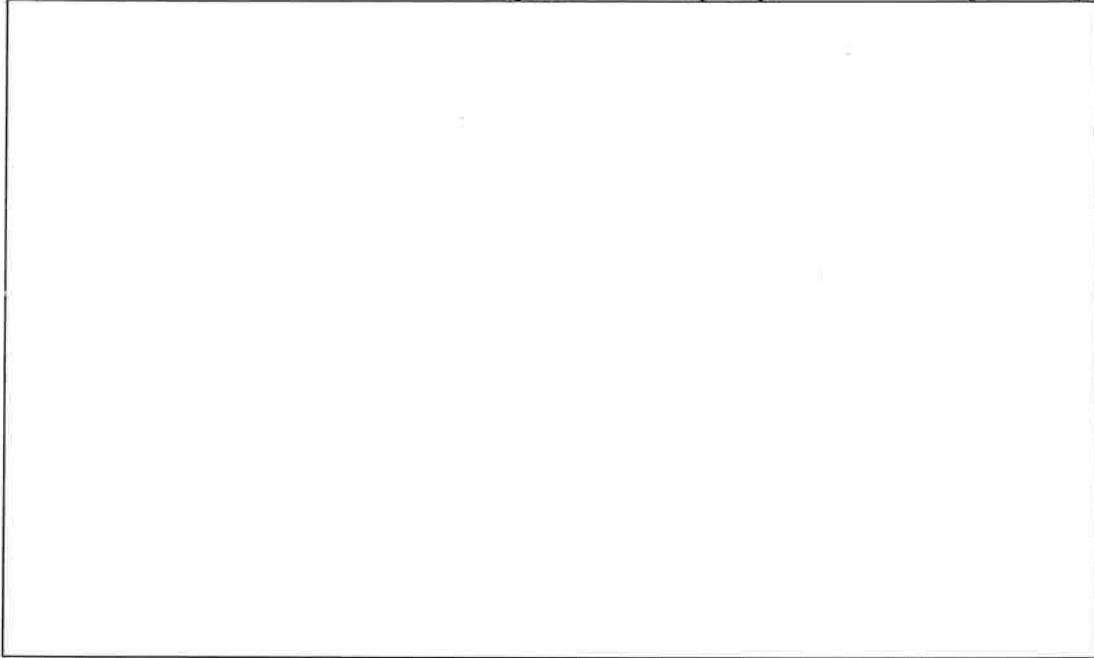
QUESTION 5

- a) Draw the simplified circuit diagram for a counter with loop sequence 0,3,7,5, 2,1.
For any unwanted state, it is considered as don't care. [8 Marks]

**Continued**

**Continued**

b) Draw a 4-bit Serial In/Parallel Out register with D flip-flop. [2 Marks]



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